

Develop a C program using fork() that creates a parent and child process. Display the Process ID (PID), Parent Process ID (PPID), and process states at different stages of execution.

```
snigdha@Snigdha:~$ nano practical3.c
snigdha@Snigdha:~$ cat practical3.c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>

int main()
{
    pid_t pid;

    pid = fork();

    if (pid < 0)
    {
        printf("Fork failed!\n");
        exit(1);
    }
    else if (pid == 0)
    {
        // Child Process
        printf("\n==== Child Process =====\n");
        printf("Child PID : %d\n", getpid());
        printf("Parent PID : %d\n", getppid());

        printf("Child is running...\n");
        sleep(40); // Keeps child alive for observation

        printf("Child process terminated.\n");
    }
    else
    {
        // Parent Process
        printf("\n==== Parent Process =====\n");
        printf("Parent PID : %d\n", getpid());
        printf("Child PID : %d\n", pid);

        printf("Parent waiting for child...\n");

        wait(NULL);

        printf("Child completed.\n");
        printf("Parent process terminated.\n");
    }

    return 0;
}
```

```
snigdha@Snigdha:~$ ./practical3
```

```
==== Parent Process =====
==== Child Process =====
Parent PID : 971
Child PID : 972
Child PID : 972
Parent waiting for child...
Parent PID : 971
Child is running...
Child process terminated.
Child completed.
Parent process terminated.
```

Design an experiment to observe process state transitions (Ready, Running, Waiting, Terminated) using Linux monitoring tools such as ps, top, and /proc. Document the observations.

```
snigdha@Snigdha: ~  
snigdha@Snigdha:~$ ps -ef | grep practical3  
snigdha      867      349  0 03:02 pts/0    00:00:00 ./practical3  
snigdha      868      867  0 03:02 pts/0    00:00:00 ./practical3  
snigdha      904      877  0 03:03 pts/2    00:00:00 grep --color=auto practic  
al3  
snigdha@Snigdha:~$ ps -o pid,ppid,state,cmd -p 868,867  
PID    PPID S CMD  
snigdha@Snigdha:~$ ps -o pid,ppid,state,cmd -p 867,868  
PID    PPID S CMD  
snigdha@Snigdha:~$ cat /proc/868/status  
cat: /proc/868/status: No such file or directory  
snigdha@Snigdha:~$ ps -o pid,ppid,state,cmd -p 971,972  
PID    PPID S CMD  
971     349 S ./practical3  
972     971 S ./practical3  
snigdha@Snigdha:~$ cat /proc/972/status  
Name: practical3  
Umask: 0022  
State: S (sleeping)  
Tgid: 972  
Ngid: 0  
Pid: 972  
PPid: 971  
TracerPid: 0  
Uid: 1000 1000 1000 1000  
Gid: 1000 1000 1000 1000  
FDSize: 64  
Groups: 4 24 27 30 46 100 1000  
NSTgid: 972  
NSpid: 972  
NSpgid: 971  
NSSid: 349  
Kthread: 0  
VmPeak: 2768 kB  
VmSize: 2768 kB  
VmLck: 0 kB  
VmPin: 0 kB  
VmHWM: 1116 kB  
VmRSS: 1116 kB
```